

CO1 - Assessment Tool 2 -Application BasedQuestions

CODE: DSA0506-QUERRY PROCESSSSING FOR DATASCIENCE

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1.Banking ATM Transaction Monitoring (10 Marks)

A bank stores ATM information in XML format and transaction logs in CSV format. Write a Python program to identify ATMs with unusually high transaction volumes and generate a monitoring report.

CODE:

```
import pandas as pd

import xml.etree.ElementTree as ET

from io import StringIO

# XML Data
xml_data = """
<ATMs>
  <ATM>
    <ATM_ID>ATM101</ATM_ID>
    <Location>Chennai</Location>
  </ATM>
  <ATM>
    <ATM_ID>ATM102</ATM_ID>
    <Location>Vijayawada</Location>
  </ATM>
  <ATM>
    <ATM_ID>ATM103</ATM_ID>
    <Location>Hyderabad</Location>
  </ATM>
</ATMs>
```

```

"""
# Read XML
root = ET.fromstring(xml_data)
atm_list = []
for atm in root.findall("ATM"):
    atm_list.append({
        "ATM_ID": atm.find("ATM_ID").text,
        "Location": atm.find("Location").text
    })
atm_df = pd.DataFrame(atm_list)
# CSV Data
csv_data = """ATM_ID,Transactions
ATM101,120
ATM102,350
ATM103,180
"""

transaction_df = pd.read_csv(StringIO(csv_data))
# Merge
result = pd.merge(atm_df, transaction_df, on="ATM_ID")
# High Transaction
result["Status"] = result["Transactions"].apply(
    lambda x: "High Transaction" if x > 200 else "Normal"
)
print("ATM Monitoring Report")
print(result)

```

OUTPUT:

```

... ATM Monitoring Report
   ATM_ID  Location  Transactions  Status
0  ATM101   Chennai          120   Normal
1  ATM102 Vijayawada          350 High Transaction
2  ATM103   Hyderabad          180   Normal

```

2.Municipal Waste Collection (10 Marks)

A municipality stores waste collection schedules in JSON format and vehicle movement records in CSV format. Write a Python program to combine the datasets and identify areas where waste collection has been delayed.

CODE:

```
import pandas as pd

import json

from io import StringIO

# JSON Data
json_data = '''

[
    {"Area":"A1","Scheduled_Time":"08:00"},
    {"Area":"A2","Scheduled_Time":"09:00"},
    {"Area":"A3","Scheduled_Time":"10:00"}
]

'''

schedule_df = pd.DataFrame(json.loads(json_data))

# CSV Data
csv_data = """Area,Actual_Time
A1,08:05
A2,09:45
A3,10:00
"""

vehicle_df = pd.read_csv(StringIO(csv_data))

# Merge
result = pd.merge(schedule_df, vehicle_df, on="Area")

# Delay Checking
result["Delayed"] = result["Scheduled_Time"] != result["Actual_Time"]

print("Waste Collection Report")
```

```
print(result)
```

OUTPUT:

```
... Waste Collection Report
   Area Scheduled_Time Actual_Time Delayed
0   A1          08:00       08:05     True
1   A2          09:00       09:45     True
2   A3          10:00       10:00    False
```